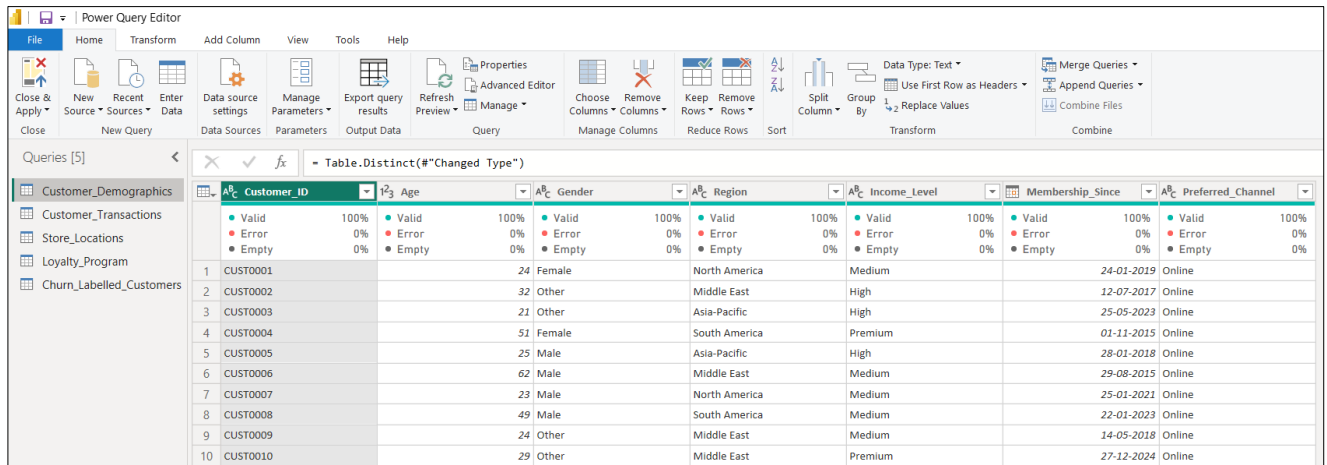


Power – BI

Retail Customer Retention Analytics – ADIDAS

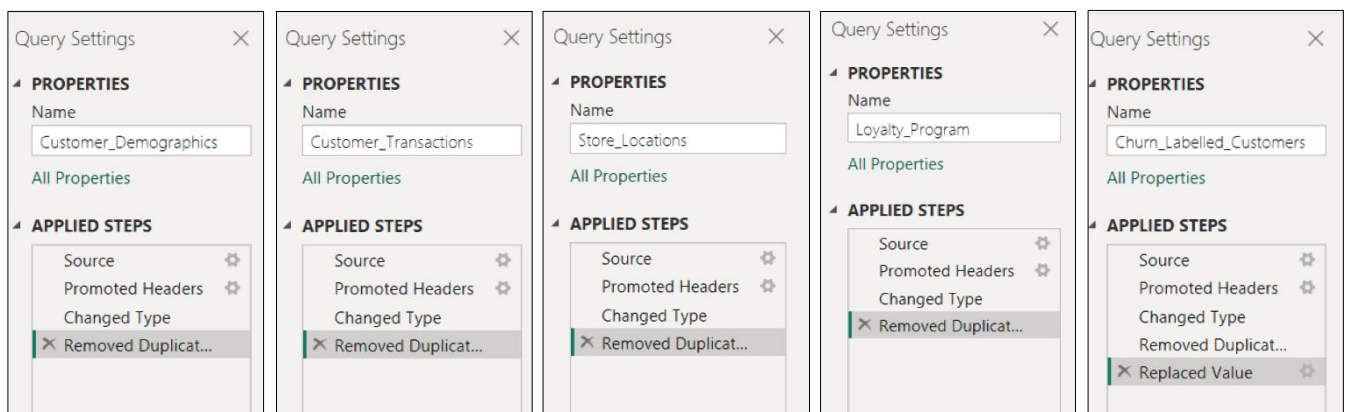
Task 1: Data Modeling & Cleaning

- Load and transform datasets in Power Query



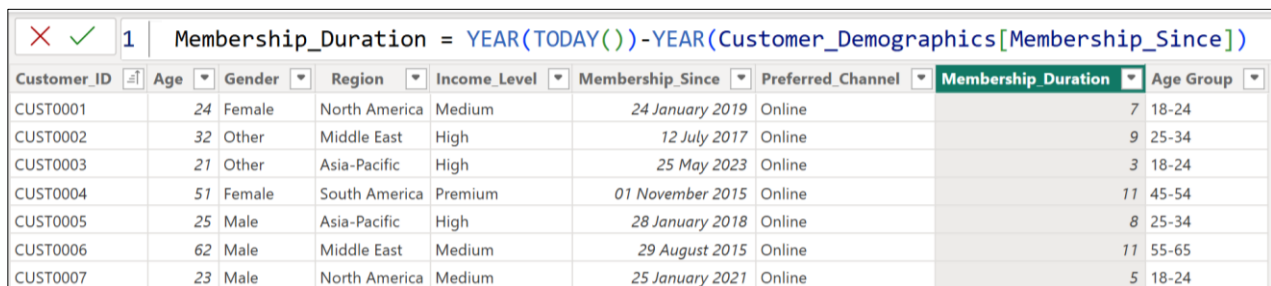
	Customer_ID	Age	Gender	Region	Income_Level	Membership_Since	Preferred_Channel
1	CUST0001	24	Female	North America	Medium	24-01-2019	Online
2	CUST0002	32	Other	Middle East	High	12-07-2017	Online
3	CUST0003	21	Other	Asia-Pacific	High	25-05-2023	Online
4	CUST0004	51	Female	South America	Premium	01-11-2015	Online
5	CUST0005	25	Male	Asia-Pacific	High	28-01-2018	Online
6	CUST0006	62	Male	Middle East	Medium	29-08-2015	Online
7	CUST0007	23	Male	North America	Medium	25-01-2021	Online
8	CUST0008	49	Male	South America	Medium	22-01-2023	Online
9	CUST0009	24	Other	Middle East	Medium	14-05-2018	Online
10	CUST0010	29	Other	Middle East	Premium	27-12-2024	Online

- Handle duplicates, missing values, and ensure correct data types



Query Settings	Query Settings	Query Settings	Query Settings	Query Settings
PROPERTIES Name: Customer_Demographics	PROPERTIES Name: Customer_Transactions	PROPERTIES Name: Store_Locations	PROPERTIES Name: Loyalty_Program	PROPERTIES Name: Churn_Labelled_Customers
APPLIED STEPS Source Promoted Headers Changed Type X Removed Duplicat...	APPLIED STEPS Source Promoted Headers Changed Type X Removed Duplicat...	APPLIED STEPS Source Promoted Headers Changed Type X Removed Duplicat...	APPLIED STEPS Source Promoted Headers Changed Type X Removed Duplicat...	APPLIED STEPS Source Promoted Headers Changed Type Removed Duplicat... X Replaced Value

- Create calculated columns:
 - Membership_Duration = Today – Membership_Since



	Customer_ID	Age	Gender	Region	Income_Level	Membership_Since	Preferred_Channel	Membership_Duration	Age Group
1	CUST0001	24	Female	North America	Medium	24 January 2019	Online	7	18-24
2	CUST0002	32	Other	Middle East	High	12 July 2017	Online	9	25-34
3	CUST0003	21	Other	Asia-Pacific	High	25 May 2023	Online	3	18-24
4	CUST0004	51	Female	South America	Premium	01 November 2015	Online	11	45-54
5	CUST0005	25	Male	Asia-Pacific	High	28 January 2018	Online	8	25-34
6	CUST0006	62	Male	Middle East	Medium	29 August 2015	Online	11	55-65
7	CUST0007	23	Male	North America	Medium	25 January 2021	Online	5	18-24

- **Extract Transaction_Year,**

1 Transaction_Year = YEAR(Customer_Transactions[Transaction_Date])

Transaction_ID	Customer_ID	Store_ID	Product_Category	Transaction Date	Amount	Promotion Applied	Transaction Year	Transaction_Month
TXN00001	CUST0157	STORE015	Footwear	27 June 2023	372.57	Yes	2023	6
TXN00002	CUST0187	STORE038	Accessories	28 July 2023	216.02	Yes	2023	7
TXN00003	CUST0175	STORE038	Apparel	24 August 2025	433.22	Yes	2025	8
TXN00004	CUST0214	STORE008	Apparel	06 August 2025	142.29	Yes	2025	8
TXN00005	CUST0047	STORE048	Footwear	28 October 2022	181.87	Yes	2022	10
TXN00006	CUST0048	STORE010	Apparel	04 November 2022	89.68	No	2022	11
TXN00007	CUST0099	STORE048	Accessories	11 January 2023	63.7	No	2023	

- **Transaction_Month**

Transaction_ID	Customer_ID	Store_ID	Product_Category	Transaction_Date	Amount	Promotion_Applied	Transaction_Year	Transaction_Month	Store_Type
TXN00001	CUST0157	STORE015	Footwear	27 June 2023	372.57	Yes	2023	6	Outlet
TXN00002	CUST0187	STORE038	Accessories	28 July 2023	216.02	Yes	2023	7	Flagship
TXN00003	CUST0175	STORE038	Apparel	24 August 2025	433.22	Yes	2025	8	Flagship
TXN00004	CUST0214	STORE008	Apparel	06 August 2025	142.29	Yes	2025	8	Outlet
TXN00005	CUST0047	STORE048	Footwear	28 October 2022	181.87	Yes	2022	10	Outlet
TXN00006	CUST0048	STORE010	Apparel	04 November 2022	89.68	No	2022	11	Franchise
TXN00007	CUST0099	STORE048	Accessories	11 January 2023	63.7	No	2023	1	Outlet

Task 2: Churn & Retention Metrics

Task 2: Churn & Retention Metrics

- Create **Churn Rate KPI** = (Churned Customers / Total Customers) * 100

Churned Customers

```
Churned Customers = CALCULATE(COUNT(Churn_Labelled_Customers[Customer_ID]),
Churn_Labelled_Customers[Churn_Flag] = 1)
```

Total Customers

1 Total Customers = `DISTINCTCOUNT(Churn_Labelled_Customers[Customer_ID])`

Churned Customers
147

Total Customers
500

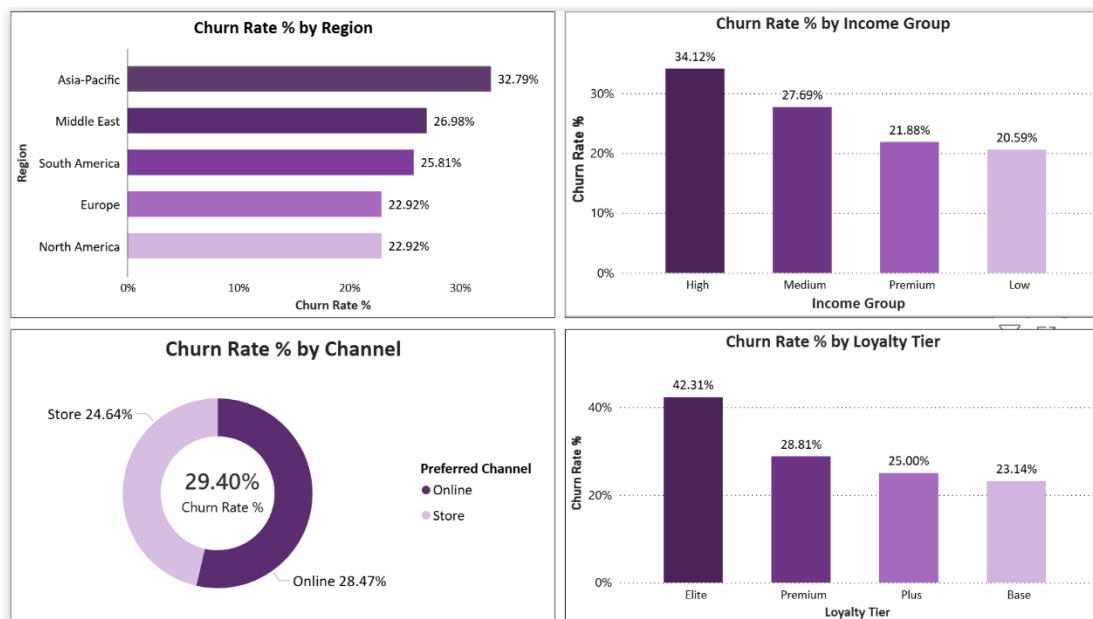
- Create **Churn Rate KPI** = (Churned Customers / Total Customers) * 100

Churn Rate % = `DIVIDE([Churned Customers],[Total Customers],0)`

Churn Rate %
29.40%

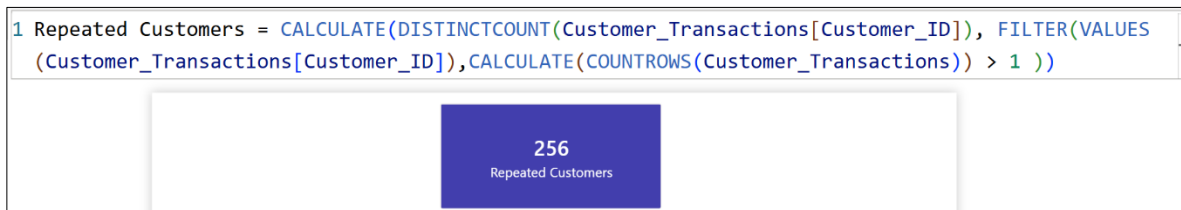
- **Visualize churn rate by:**

- **Region** ○ **Income Group** ○ **Channel (Store/Online)** ○ **Loyalty Tier**

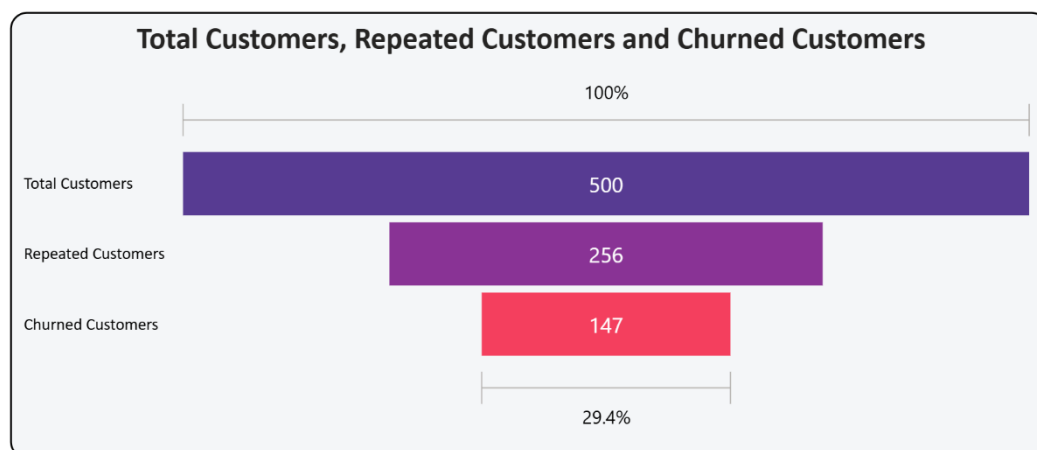


- **Funnel Chart:** Total Customers → Repeat Customers → Churned

Repeat Customers



Total Customers → Repeat Customers → Churned



Task 3: Repeat Purchase Analysis

Task 3: Repeat Purchase Analysis

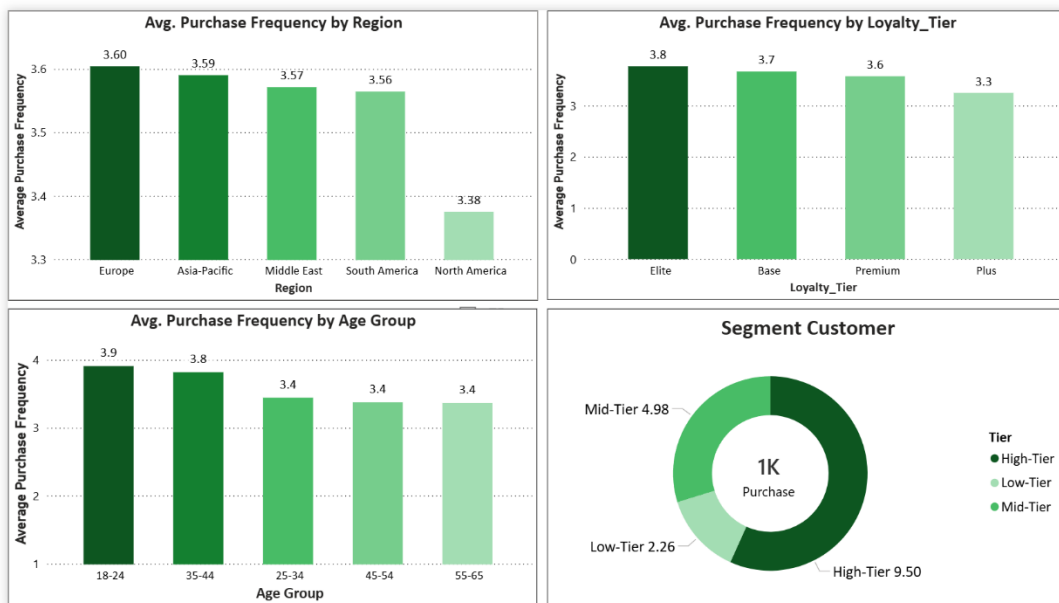
- Segment customers:
 - Low-Tier:** 0–3 purchases
 - Mid-Tier:** 4–8 purchases
 - High-Tier:** 9+ purchases

To calculate **purchase count**, use power Query (Group by feature)

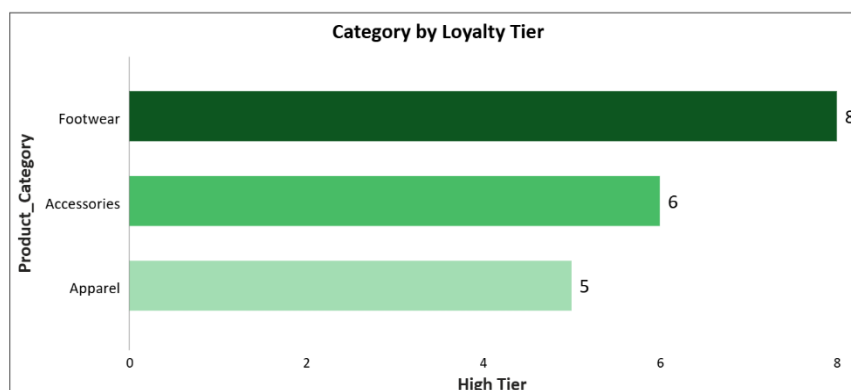
To calculate **Tier**, use power Query (Conditional Column)

Segment Customer		
Customer_ID	Purchase Count	Tier
CUST0001	7	Mid-Tier
CUST0002	2	Low-Tier
CUST0003	2	Low-Tier
CUST0004	7	Mid-Tier
CUST0005	5	Mid-Tier
CUST0007	4	Mid-Tier
CUST0008	3	Low-Tier
CUST0009	5	Mid-Tier
CUST0010	2	Low-Tier
CUST0011	7	Mid-Tier
CUST0012	5	Mid-Tier
CUST0014	6	Mid-Tier
CUST0015	1	Low-Tier

- Compare avg. purchase frequency by **Region, Age Group, Loyalty Tier**



- Identify most purchased product categories by loyal customers



Task 4: Promotion & Loyalty Impact

Task 4: Promotion & Loyalty Impact

- % of transactions with promotion applied

Transactions

```
1 Total Transction = COUNT(Customer_Transactions[Transaction_ID])
```

1K
Total Transction

with promotion applied

```
1 with promotion applied = CALCULATE(COUNT(Customer_Transactions[Transaction_ID]), Customer_Transactions[Promotion_Applied] = "Yes")
```

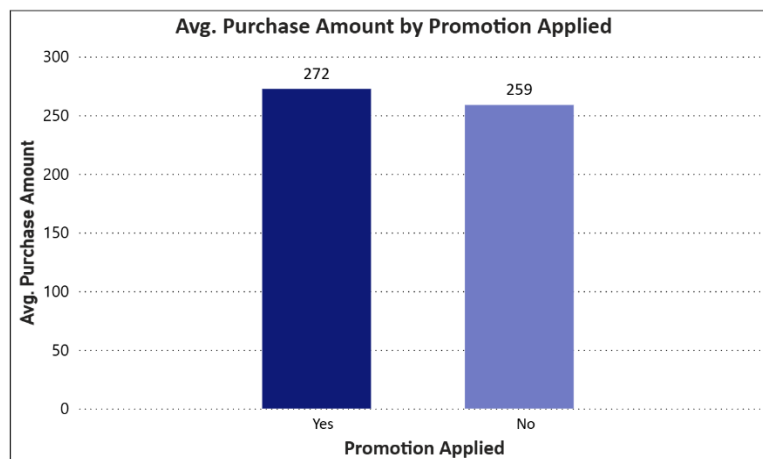
527
with promotion applied

- % of transactions with promotion applied

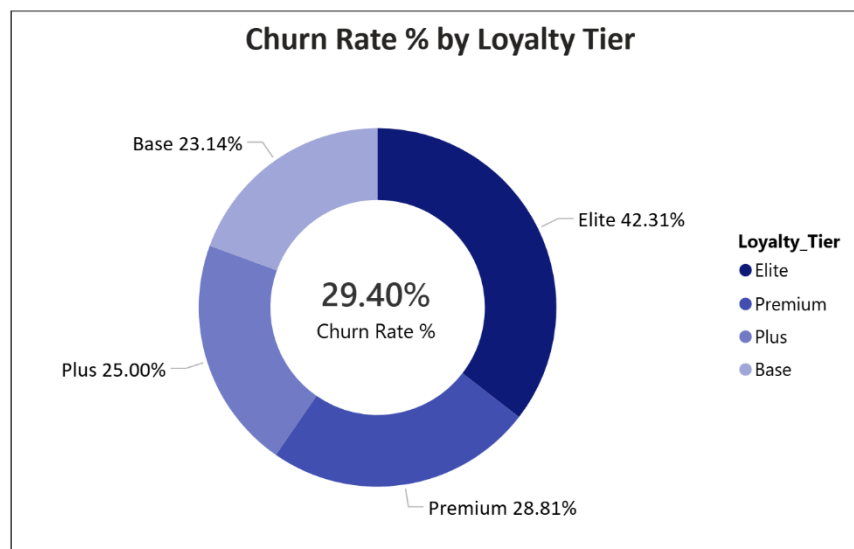
```
(%) of Transction with promotion applied = DIVIDE([with promotion applied],[Total Transction],0)
```

52.70%
(%) of Transction with promotion applied

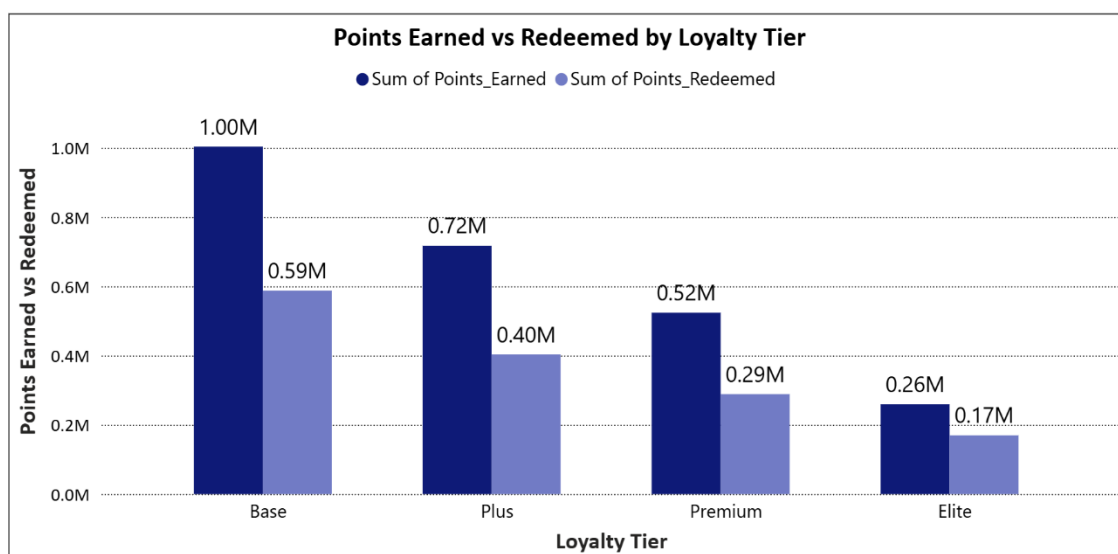
- Compare **avg. purchase amount with vs without promotions**



- Churn rate across **loyalty tiers**



- Points Earned vs Redeemed by Tier (clustered column chart)



- Recommendations to improve redemption & retention**

- Simplify the redemption process to make it easy for customers to use points.
- Increase awareness about loyalty points through notifications and reminders.
- Offer attractive discounts and benefits during redemption to encourage usage.
- Give personalized offers based on customer preferences.
- Send alerts for points expiry to create urgency for redemption .
- Provide promotions, rewards, and engagement offers to improve customer retention.

Task 5: Store & Channel Performance vs Retention

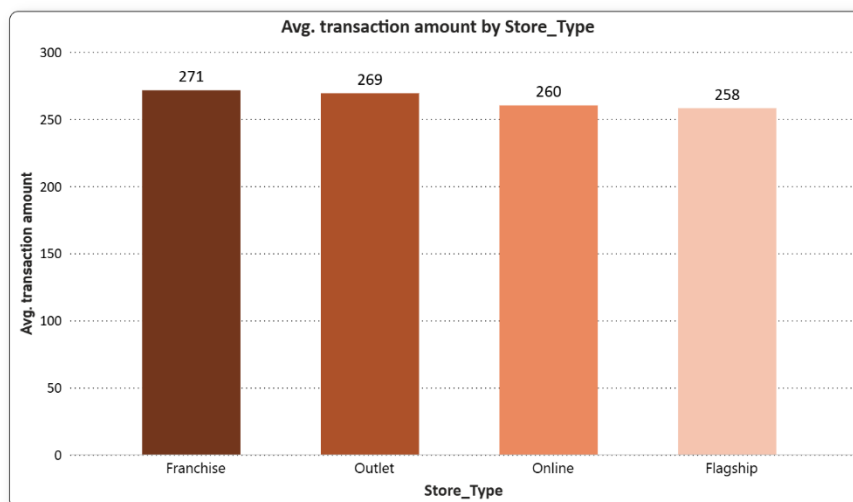
Task 5: Store & Channel Performance vs Retention

- Merge store data with transactions

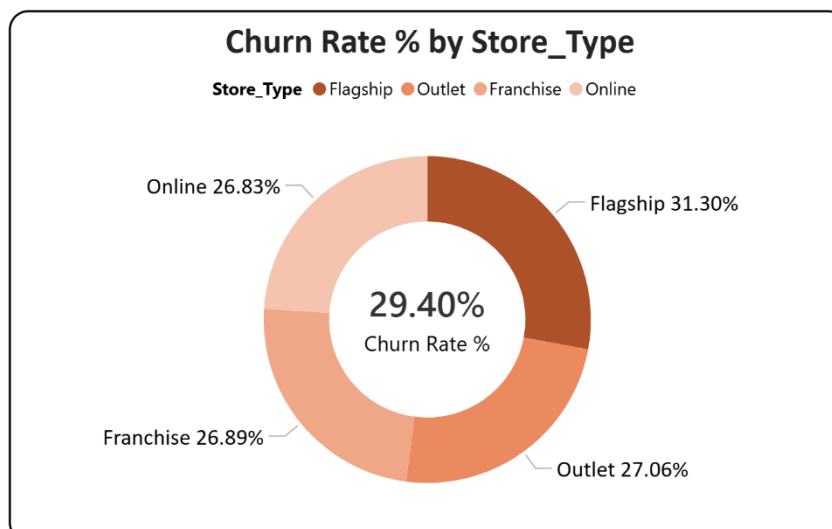
[Use Merge Queries]

Transaction_ID	Customer_ID	Store_ID	Product_Category	Transaction_Date	Amount	Promotion_Applied	Store_Type	Region	Opening_Year
TXN00001	CUST0157	STORE015	Footwear	27 June 2023	372.57	Yes	Outlet	Asia-Pacific	2013
TXN00002	CUST0187	STORE038	Accessories	28 July 2023	216.02	Yes	Flagship	South America	2013
TXN00003	CUST0175	STORE038	Apparel	24 August 2025	433.22	Yes	Flagship	South America	2013
TXN00004	CUST0214	STORE008	Apparel	06 August 2025	142.29	Yes	Outlet	Middle East	2000
TXN00005	CUST0047	STORE048	Footwear	28 October 2022	181.87	Yes	Outlet	North America	2003
TXN00006	CUST0048	STORE010	Apparel	04 November 2022	89.68	No	Franchise	Europe	2001
TXN00007	CUST0099	STORE048	Accessories	11 January 2023	63.7	No	Outlet	North America	2003
TXN00008	CUST0026	STORE004	Footwear	23 June 2024	246.53	No	Franchise	Europe	2013
TXN00009	CUST0089	STORE006	Footwear	18 August 2023	167.57	Yes	Outlet	Europe	2015
TXN00010	CUST0071	STORE033	Accessories	13 January 2023	28.76	No	Online	Europe	2010
TXN00011	CUST0239	STORE034	Footwear	10 November 2024	186.47	Yes	Online	Europe	2006
TXN00012	CUST0184	STORE018	Apparel	16 January 2023	411	Yes	Outlet	South America	2015
TXN00013	CUST0140	STORE033	Accessories	20 June 2023	189.68	Yes	Online	Europe	2010
TXN00014	CUST0162	STORE012	Apparel	03 April 2025	87.92	Yes	Outlet	Middle East	2016
TXN00015	CUST0152	STORE019	Accessories	22 January 2024	80.96	No	Flagship	North America	2007

- Visualize:
 - Avg. transaction amount by **Store Type**



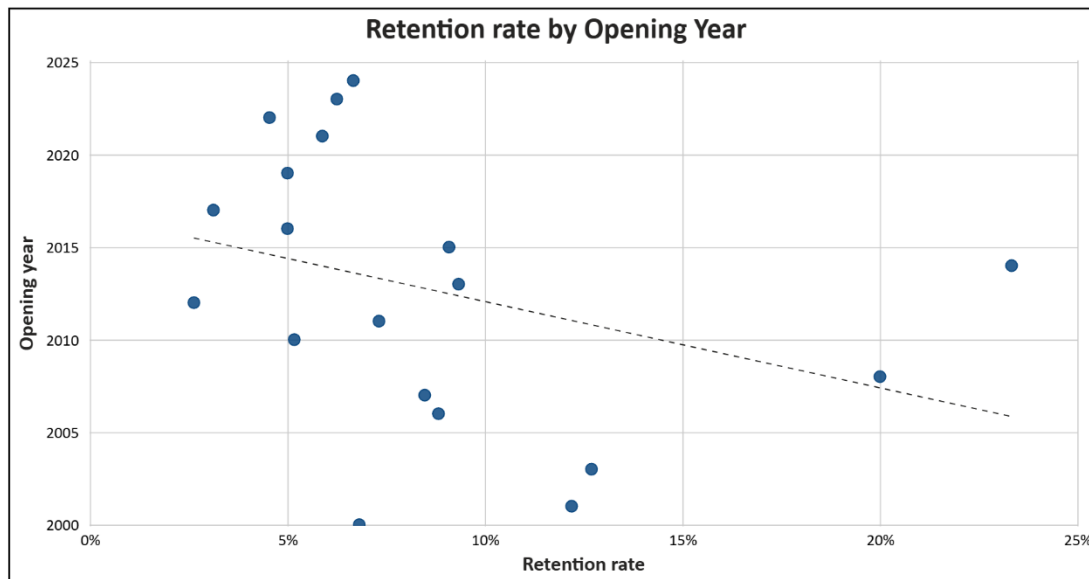
- Churn rate by store type



- Correlation between **store opening year & retention**

```
1 Retention rate = DIVIDE([Repeated Customers],[Total Customers],0)
```

Retention rate
51.20%



- The scatter plot shows a slight negative relationship between store opening year and retention rate. This means that as the store opening year increases (newer stores), the retention rate tends to be slightly better. On the other hand, older stores show comparatively lower retention rates.
- Adidas should focus on improving older stores by upgrading store design, improving customer service, and increasing engagement activities. This will help in improving retention rates in those stores.

Task 6: Customer Lifetime Value (CLV) Analysis

Task 6: Customer Lifetime Value (CLV) Analysis

- CLV = Total Amount Spent / Membership Duration (Years)

Total Amount Spent

Total Amount Spent = `SUM(Customer_Transactions[Amount])`

265.98K

Total Amount Spent

Membership Duration (Years)

Membership Duration (Years) = `AVERAGE(Customer_Demographics[Membership_Duration])`

5.83

Membership Duration (Years)

CLV = Total Amount Spent / Membership Duration (Years)

1 CLV = `DIVIDE([Total Amount Spent],[Membership Duration (Years)],0)`

45.60K

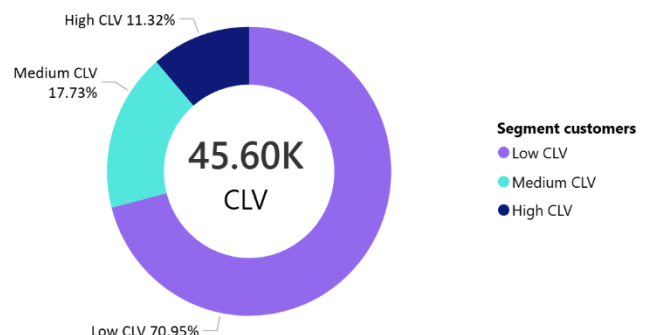
CLV

- Segment customers into Low, Medium, High CLV

1 Segment customers = `IF([CLV]<=800,"Low CLV",IF([CLV]<=1300,"Medium CLV","High CLV"))`

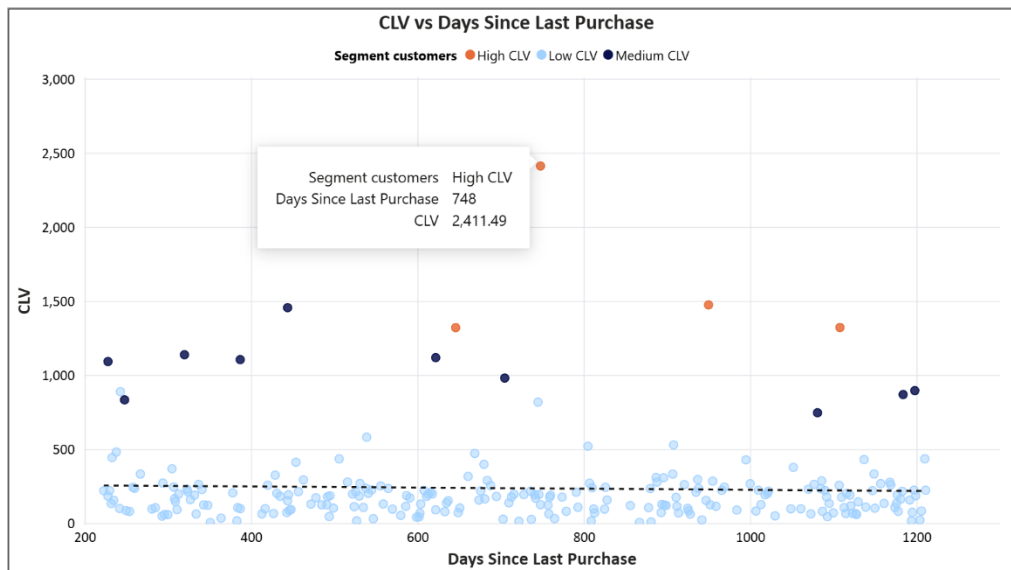
Customer_ID	Membership_Since	Membership_Duration	Sum of Amount	CLV	Segment customers
CUST0001	24 January 2019	7	1,000.16	142.88	Low CLV
CUST0002	12 July 2017	9	925.46	102.83	Low CLV
CUST0003	25 May 2023	3	877.50	292.50	Low CLV
CUST0004	01 November 2015	11	1,682.93	152.99	Low CLV
CUST0005	28 January 2018	8	1,303.94	162.99	Low CLV
CUST0007	25 January 2021	5	1,169.90	233.98	Low CLV
CUST0008	22 January 2023	3	941.66	313.89	Low CLV
CUST0009	14 May 2018	8	1,680.87	210.11	Low CLV
CUST0010	27 December 2024	2	372.92	186.46	Low CLV
CUST0011	11 October 2020	6	1,719.74	286.62	Low CLV
CUST0012	27 May 2023	3	1,643.50	547.83	Low CLV
CUST0014	26 August 2024	2	1,600.23	800.12	Medium CLV
CUST0015	06 February 2022	4	483.97	120.99	Low CLV
CUST0016	10 September 2019	7	1,102.05	157.44	Low CLV
CUST0017	06 February 2018	8	2,855.41	356.93	Low CLV
CUST0018	03 June 2023	3	1,587.99	529.33	Low CLV
CUST0019	25 January 2018	8	687.72	85.97	Low CLV
CUST0020	16 May 2017	9	1,861.65	206.85	Low CLV

CLV by Segment customers



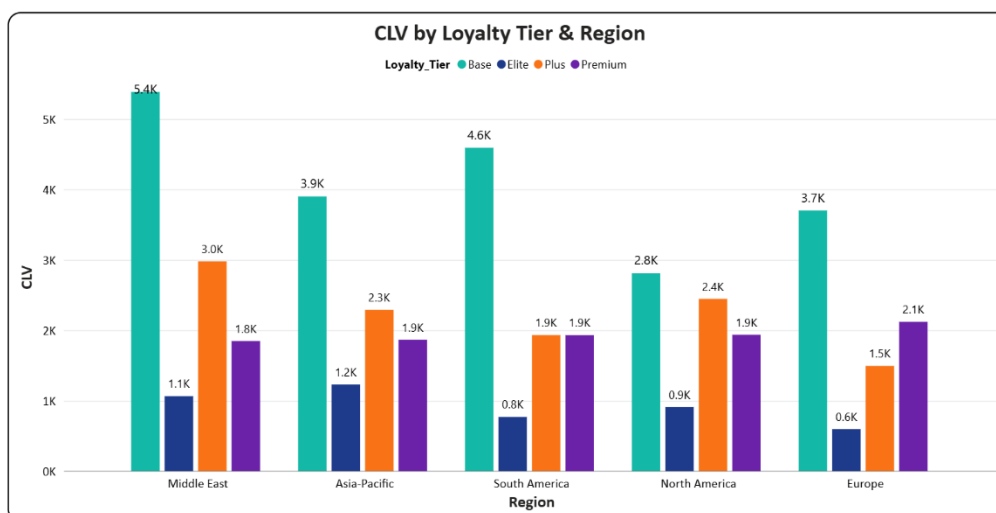
○ CLV vs Days Since Last Purchase

1 Days Since Last Purchase = `DATEDIFF(Churn_Labelled_Customers[Last_Purchase_Date], TODAY(), DAY)`



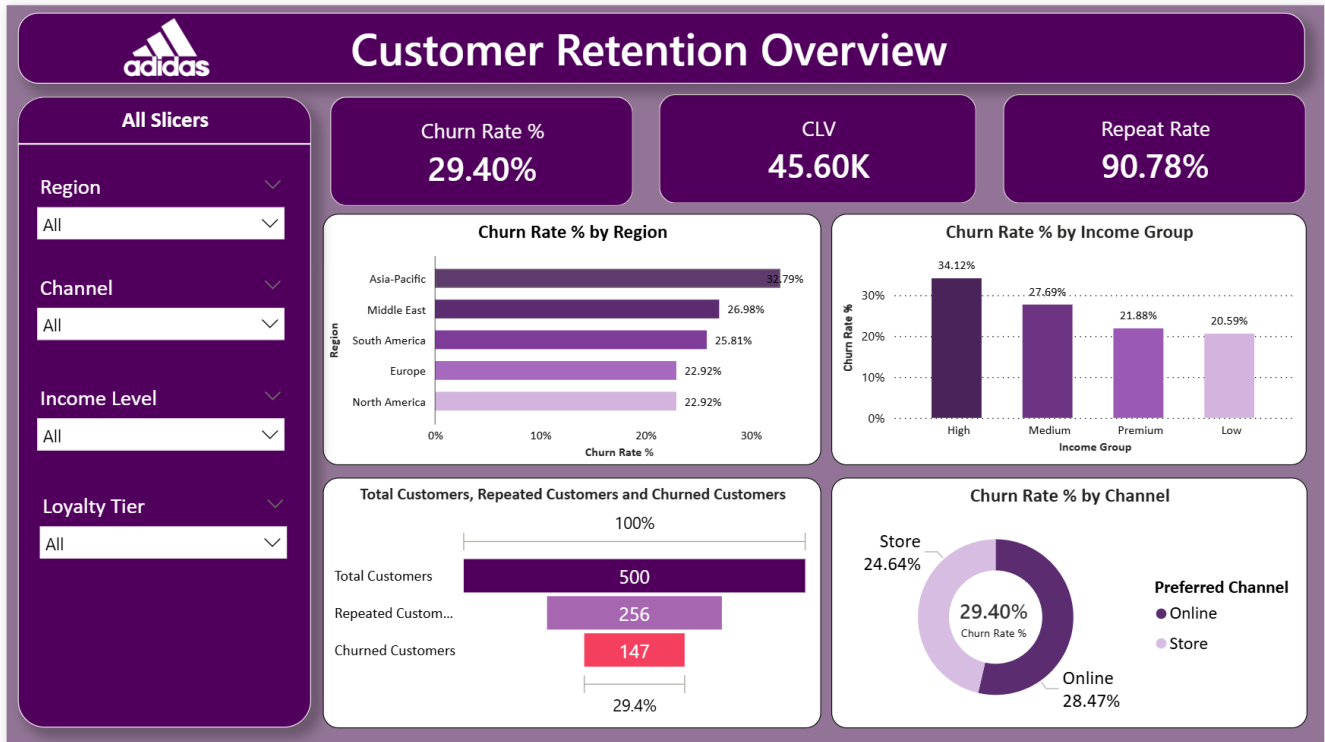
- The scatter plot shows a weak negative relationship between CLV and days since last purchase. This means that as the number of days increases (customers not purchasing for a long time), their CLV tends to be lower. Customers who purchase recently usually have higher CLV.
- Adidas should target inactive customers by sending reminders, personalized offers, and discounts to encourage repeat purchases. At the same time, high CLV customers should be retained through loyalty rewards and special benefits to maintain their engagement.

○ CLV by Loyalty Tier & Region

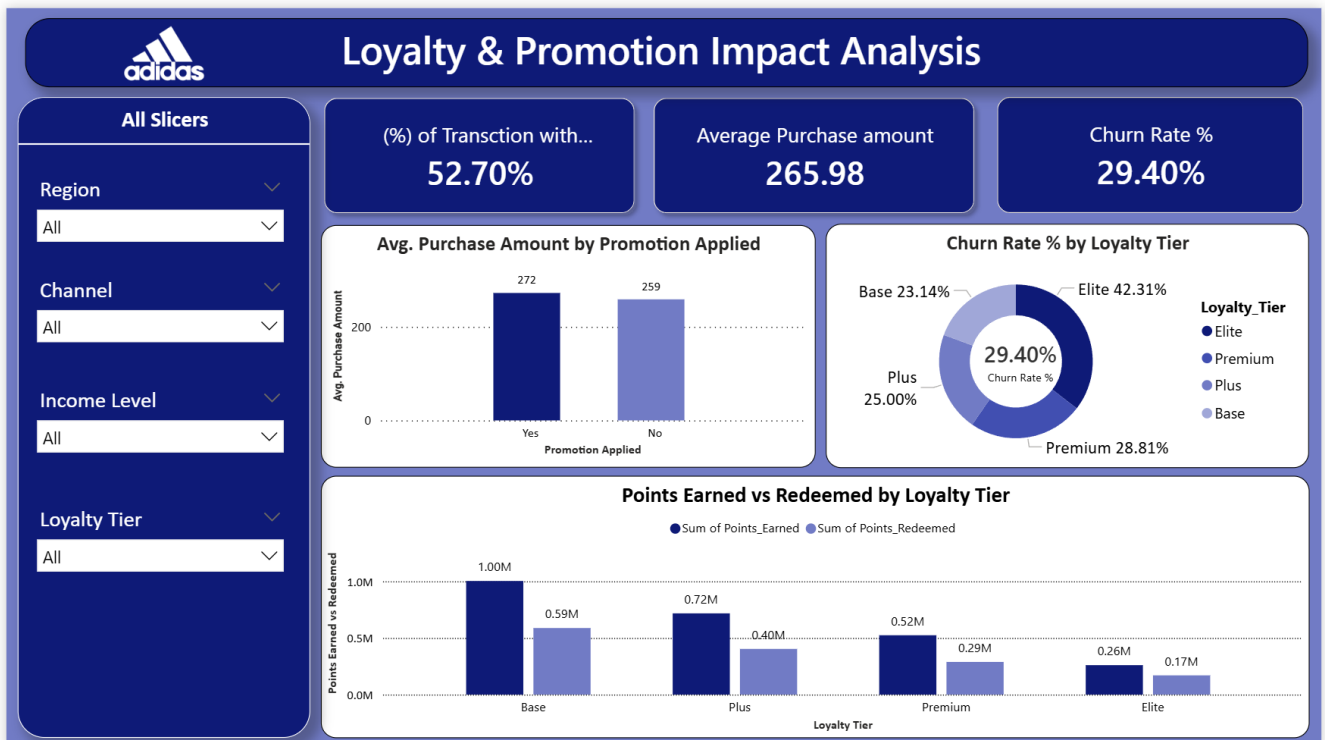


Task 7: Final Dashboard & Executive Summary

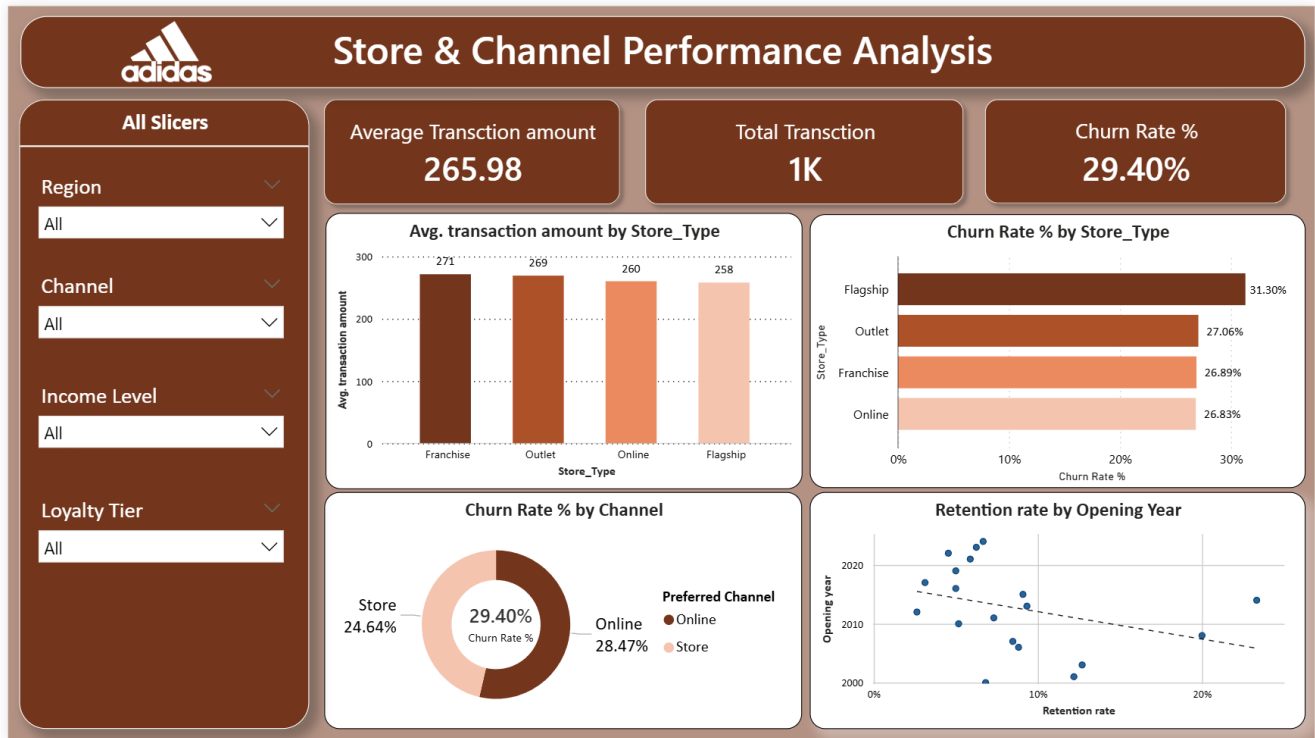
- Multi-page Power BI Report:
 - Page 1: KPIs (Churn, CLV, Repeat Rate)



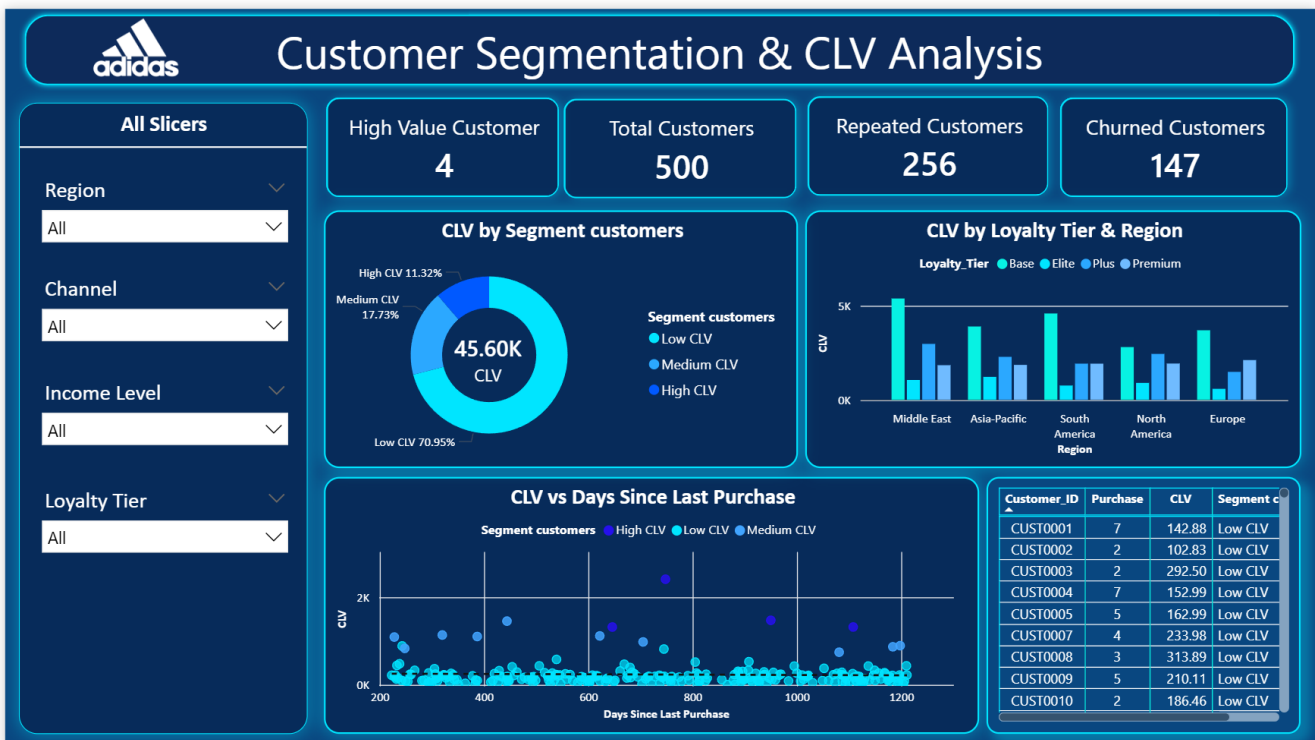
- Page 2: Loyalty & Promotion Impact



○ Page 3: Store/Channel Insights



○ Page 4: Segmentation (Churned, Repeat, High-Value)



Top 3 recommendations for Adidas:

1. Which customers to prioritize for retention?

- Adidas should prioritize high-value (high CLV) and loyal (repeat/Elite) customers, as they contribute the most revenue, and also focus on at-risk customers who have not purchased recently, as retaining them will help reduce churn and improve long-term customer value.

2. Which channels are underperforming?

- Online channel is underperforming due to higher churn. Improve online channel by providing faster delivery, better customer support, easy returns, and attractive offers to reduce churn.
- within stores types are also underperforming due to relatively higher churn compared to other store types, indicating issues in customer engagement and low awareness. It should be improved by enhancing in-store experience, personalized service, and customer interaction.

3. How to strengthen loyalty program engagement ?

- Adidas should strengthen its loyalty program by simplifying redemption, offering attractive and personalized rewards, and increasing awareness, which will encourage customers to engage more and improve retention

Task 8: Video explanation: Expressing the finding and actionable insights

Video Link

[Click To Watch Video](#)